

Vamshi Kodipaka

FCT PORTUGAL · PHD CANDIDATE

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“Be the change that you want to see in the world.” - Mahatma Gandhi

Summary

PhD Researcher in aerial robotics, computer vision, and embodied AI specializing in transformer-based aerial perception, active sensing, informative path planning, and autonomous UAV systems for search-and-rescue. Developed lightweight perception models for edge deployment and integrated detection with decision-making under uncertainty using reinforcement learning and POMDPs. Experienced in three national projects and academic publishing.

Work Experience

Institute of Systems and Robotics, University of Coimbra

Coimbra, Portugal

RESEARCHER

Sep. 2022 - Present

- Goal is to develop and integrate the novel informative planning and detection system for drone-based missing person searches in the wilderness.
- Proposed Synaptics-guided Hybrid YOLO-Transformers and achieved the robustness over 30% for tiny and small aerial person detection.
- Working on UAV informative path planning approaches for effective search, deployment on ROS + NVIDIA Jetson with field experiments.

University of Coimbra

Coimbra, Portugal

RESEARCH FELLOW

Mar. 2022 - Jun. 2022

- Developed artificial perception mechanisms supporting forestry ground robotic operations for land maintenance and clearance.
- Extended work to 2D Dynamic Entity Tracking with Multispectral Images to dockers, dataset labelling, and Stereo-3D-Reconstruction in ROS.
- Project: Safety, Exploration and Maintenance of Forests with the Integration of Ecological Robotics (SemFire)
- Supervisors: Prof. David Portugal (ISR-UC), Prof. João Filipe Ferreira (ISR-UC/NTU, UK)

Laboratoire d'Analyse et d'Architecture des Systèmes, LAAS - CNRS

Toulouse, France

RESEARCH INTERN

Feb. 2020 - Jul. 2020

- Developed a vision-based integrated system on a quadrotor and a rover, and develop algorithms to track another moving quadrotor outdoors.
- Designed a novel approach for Occlusion Avoidance by Mathematical Modelling, surveyed literature for learning approaches in tracking, and benchmarked Convolutional Neural Networks with PC configurations, CPU and GPUs.
- Project: Control of Multi-Robot Systems for Observing Dynamical Phenomena (MuRoPhen)

Education

University of Coimbra

Coimbra, Portugal

DOCTOR OF PHILOSOPHY (PHD) IN ELECTRICAL ENGINEERING AND INTELLIGENT SYSTEMS

Sep. 2022 - Present

- Thesis: Navigation Systems for Search and Rescue Operations with Drones
- Supervisors: Prof. Lino Marques, Prof. Rui Cortesao, Prof. Helder Araujo
- Courses: Machine Learning for Intelligent Systems, Methodologies of Mobile Robotics
- Dataset releases: ISR Fluvial Forest Drone Data 2023

Université de Bourgogne

Le Creusot, France

MASTER OF SCIENCE IN COMPUTER VISION AND ROBOTICS (VIBOT)

Sep. 2018 - Sep. 2020

- Thesis: Vision-based Multi-robot Target Tracking with Aerial and Ground Robots
- Supervisors: Prof. Antonio Franchi (University of Twente, Netherlands) and Prof. Daniel Sidobre (LAAS-CNRS, France)
- Courses: Visual Perception, Machine Learning, Probabilistic Robotics, Autonomous Robotics, Software Engineering, Applied Mathematics
- Grade: 14.0/20.0

Kakatiya University

Warangal, India

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND INSTRUMENTATION ENGINEERING

Sep. 2012 - Sep. 2016

- Thesis: Performance evaluation of Various QRS Detection Algorithms
- Supervisors: Prof. Shashikanth Boorla (KITSW) and Prof. Venu Madhav Kottur (KITSW)
- Courses: Image Processing, Signals, Computer Architecture, Electronic Devices, Measurements, Sensors, Process Control, Analog/Digital IC, Control Systems, Mechatronics, Mathematics, Physics
- Grade: 77.6 %

Publications

AUTHORS: V KODIPAKA, LF MARQUES, RPD CORTESÃO, HJ ARAÚJO (VAMSHI FIRST AUTHORED ALL)

RAS, 2025 (In Review) TPH-YOLOv7t: A Boosted YOLO Transformer Prediction Head for Search and Rescue with Drones

ROBOT: 6th Iberian Robotics Conference, 2023 APH-YOLOv7t: A YOLO Attention Prediction Head for Search and Rescue with Drones

BMVC, 2026 (In Review) Synaptra: Composable Synaptic-Routed Attention for Aerial Person Detection

WACV, 2027 (In Review) Ingenious: Sparse-Routed Synaptic Attention via Low-Rank Linear Projections for Aerial Person Detection

2026 (to Submit) Retention-YOLO: An Ultraboost YOLO Retention Prediction Head for Search and Rescue with Drones

Projects & Training

PROJECTS

- ROS Project: Autonomous Navigation and Tag Detection for IoT Applications
- Development of IPCV Toolbox for Camera Simulating System
- Tracking Multiple Objects in Surveillance Cameras and Videos
- Extension for Harris Operator for Interest Point Detection on 3D Meshes in C++
- Performance evaluation of Various Image Denoising Algorithms
- Visual Inspection in Soft Drink Bottle Plant; Face Recognition System using PCA
- Music Genre Classification; Visual Servoing from Lines, Points, and Image

TRAINING

- Participated in Ocean Lander Challenge at sponsored **MIT Portugal's Marine Robotics Summer School**, in Azores Islands (2024)
- Deep Learning Specialization, Aerial Robotics and Basics of Reinforcement Learning – Coursera (2020)
- Participated in LAAS-MBZIRC Seminar – International Robotic Challenge and Experiences, Abu Dhabi (2020)
- MOOCs: Elements of AI - University of Helsinki (2018), and Geospatial Applications for Disaster Risk Management - ISRO/UNOOSA (2021)
- International Astronomical Asteroid Search Collaboration, Space App Challenge Paris – NASA (2019/21)
- Industrial Visits: NTPC-Ramgundam, Indian Railways-Kazipet, and ATI-EPI Hyderabad (2012-16)

VOLUNTEERING

- Reviewer for 11th ECMR (2023), 7th Iberian Robotics Conference (2024), and 18th International Symposium on DARS (2026)
- As a member of KITS Warangal IEEE Student Branch, participated in numerous events of Warangal section (2014-16)
- Participant: WHO - COVID'19, Shiksha Shilpi's Aahar Kranti, Creating in Public Space FAI-AR, RRIITOD (Rehabilitation)
- Donated 2100 euros: UNICEF-Portugal, UNFPA, UN-WFP, WHO, ISKCON, Isha Foundation, No Baby Blisters; Blood Donation
- As an Altruist and Care-giver: donated 19,500 euros to two kids for their education

Skills

Software Python, C, C++, Java, MATLAB, Git

Libraries PyTorch, Tensorflow, ROS, Gazebo, OpenCV, Cuda

Others Linux, Oracle, LaTeX, NVIDIA Jetson Orin Nano

Languages English, Hindi, Telugu, Portuguese (Elementary)

Honors & Awards

2022	Scholarship , from FCT-Portugal for four year PhD research	Portugal
2022	Scholarship , from FCT - University of Coimbra for four months	Portugal
2020	Stipend , LAAS-CNRS Intern gratification for six months	France
2018	Scholarship , TS OBC Mahatma Jyothiba Phule Overseas Vidhyanidhi for Full M.Sc. studies	India
2016	1st Place , Best Project Award of the Year – EIE, KITSW	Warangal, India
2012	45% Fee Reimbursement , for four-year B.Tech studies	Warangal, India
2010	Fully Funded , Free Seat for two years of College	Warangal, India